Costco Basket Size Prediction

XGboost + SHAP model

- This model was built to predict Costco members' holiday spending using prior behavior and membership characteristics.

```
# Importing Libraries
import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

from xgboost import XGBRegressor
import xgboost as xgb
import shap

from google.colab import files
import io

import matplotlib.pyplot as plt
import seaborn as sns
```

Exploratory Data Analysis

- I created a synthetic dataset with 5,000 members and 14 features; such as: spending trends, visit frequency, department spending, tenure, and promo usage.

```
# File picker dialog to select the file
uploaded = files.upload()
```

Choose Files Costco_sim…ed_data.csv
Costco_simulated_data.csv(text/csv) - 1143923 bytes, last modified: 12/23/2025 - 100% done
Saving Costco_simulated_data.csv to Costco_simulated_data (2).csv

```
# Uploading the data
filename = list(uploaded.keys())[0]
costco = pd.read_csv(io.BytesIO(uploaded[filename]))
```

```
print(f"Successfully loaded '{filename}' into a DataFrame.")
```

```
Successfully loaded 'Costco_simulated_data (2).csv' into a DataFrame.
```

```
# Looking at the first 5 rows of data
display(costco.head())
```

	member_ID	rolling_12mo_spend	rolling_12mo_visit	rolling_12mo_avg_basket	r
0	1	6180.845853	28	220.744495	
1	2	3828.550871	19	201.502677	
2	3	8010.783410	21	381.465877	
3	4	5652.722854	18	314.040159	
4	5	7650.925964	24	318.788582	

```
# Data structure
costco.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 16 columns):
 #   Column                          Non-Null Count  Dtype
---  ------                          --------------  -----
 0   member_ID                       5000 non-null   int64
 1   rolling_12mo_spend              5000 non-null   float64
 2   rolling_12mo_visit              5000 non-null   int64
 3   rolling_12mo_avg_basket         5000 non-null   float64
 4   rolling_24mo_spend              5000 non-null   float64
 5   pct_spend_grocery_consumables   5000 non-null   float64
 6   pct_spend_pharmacy_health       5000 non-null   float64
 7   pct_spend_home_hardware         5000 non-null   float64
 8   pct_spend_auto_sporting         5000 non-null   float64
 9   pct_spend_general_merch         5000 non-null   float64
 10  member_tenure_years             5000 non-null   float64
 11  return_rate                     5000 non-null   float64
 12  executive_flag                  5000 non-null   int64
 13  weekend_shopping_ratio          5000 non-null   float64
 14  promo_usage_rate                5000 non-null   float64
 15  holiday_spend                   5000 non-null   float64
dtypes: float64(13), int64(3)
memory usage: 625.1 KB
```

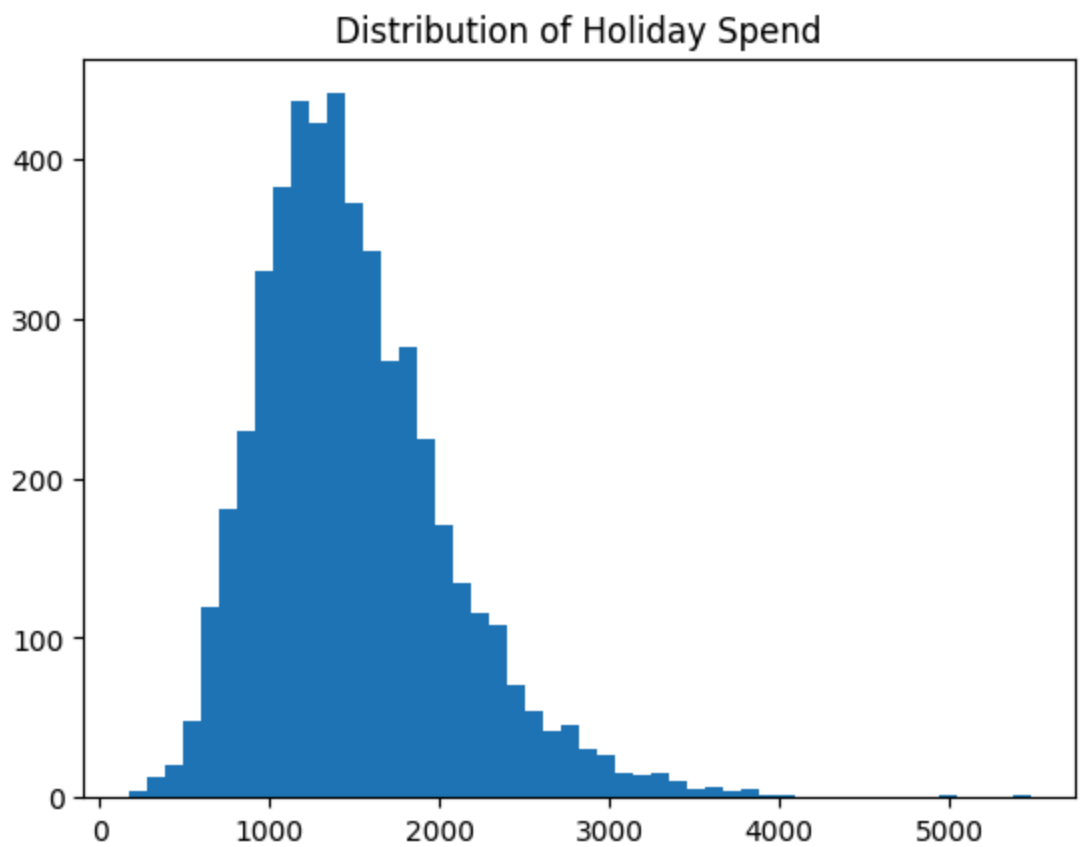
Defining Variables

```
# The target variable
target = 'holiday_spend'

x =  costco.drop(columns= [target, "member_ID"])
y = costco[target]
```

Target Distribution

- The target distribution is right skewed.

```
# Cehcking the shape of the target variable's distribution
plt.hist(y, bins=50)
plt.title("Distribution of Holiday Spend")
plt.show()

print("Skew:", y.skew())
```




```
Skew: 0.9730081766087808
```

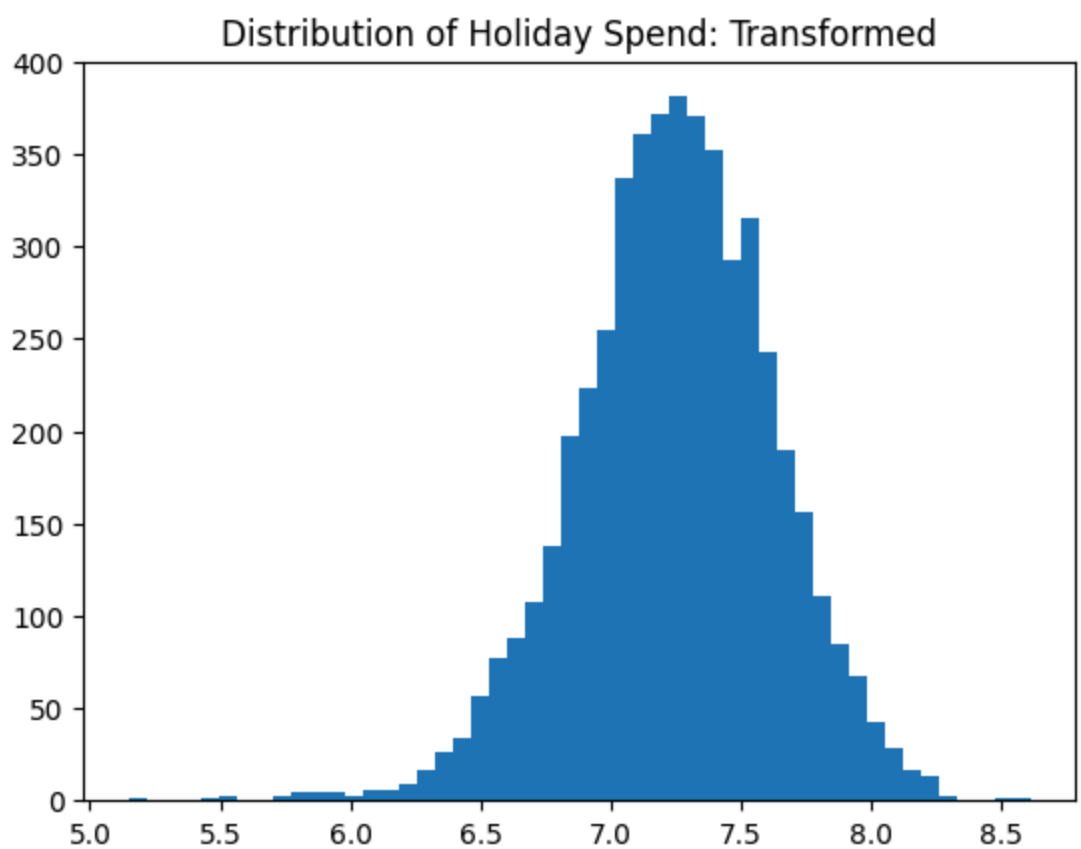
```
print(f"\nTarget variable: {target}")
print(f"Features: {list(x.columns)}")

# Check target distribution
print(f"\nTarget Statistics:")
```

```
print(f"Mean: ${y.mean():,.2f}")
print(f"Std:  ${y.std():,.2f}")
print(f"Min:  ${y.min():,.2f}")
print(f"Max:  ${y.max():,.2f}")
```

```
Target variable: holiday_spend
Features: ['rolling_12mo_spend', 'rolling_12mo_visit', 'rolling_12mo_avg_basket'

Target Statistics:
Mean: $1,501.20
Std:  $566.50
Min:  $171.04
Max:  $5,475.83
```

```
# Applying Y transformation on the Target variable
y_log = np.log1p(y)
```

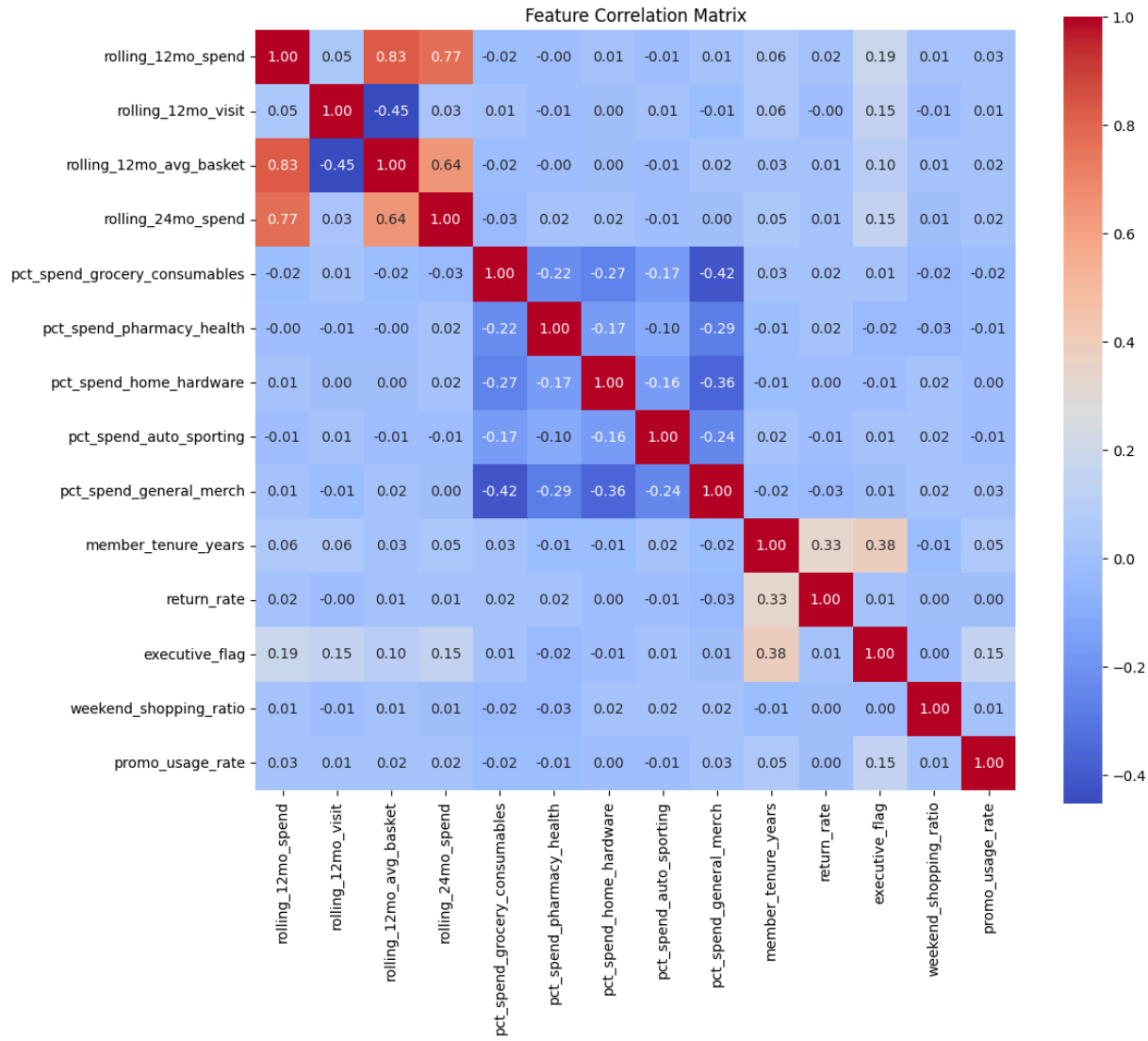
```
# Cehcking the shape of the target variable's distribution
plt.hist(y_log, bins=50)
plt.title("Distribution of Holiday Spend: Transformed")
plt.show()

print("Skew:", y.skew())
```




```
Skew: 0.9730081766087808
```

Checking for Multicollinearity

```
# Creating the correlation Matrix
corr_matrix = x.corr()

# Heatmap
plt.figure(figsize =(12, 10))
sns.heatmap(corr_matrix, annot=True, fmt=".2f", cmap="coolwarm", square=True)
plt.title("Feature Correlation Matrix")
plt.show()

# List of highly correlated pairs
threshold = 0.8  # set a threshold for "high correlation"
high_corr = []

for i in range(len(corr_matrix.columns)):
    for j in range(i+1, len(corr_matrix.columns)):
        if abs(corr_matrix.iloc[i, j]) > threshold:
            high_corr.append(
                (corr_matrix.columns[i], corr_matrix.columns[j], corr_matrix.il
            )

print("Highly correlated feature pairs (>|0.8|):")
for f1, f2, val in high_corr:
    print(f"{f1} ↔ {f2} : {val:.2f}")
```



Splitting the data

```
print("\n" + "="*60)
print("DATA SPLITTING")
print("="*60)

# Train/Test split
x_train, x_test, y_train, y_test = train_test_split(
    x, y_log,
    test_size=0.2,
    random_state=42
    )
# Further split for validation
x_train_split, x_val, y_train_split, y_val = train_test_split(
    x_train, y_train,
    test_size=0.2,
    random_state=42
    )
```

```
print(f"Training set:       {x_train.shape[0]:,} samples")
print(f"  - For training:   {x_train_split.shape[0]:,} samples")
print(f"  - For validation: {x_val.shape[0]:,} samples")
print(f"Test set:           {x_test.shape[0]:,} samples")
```

```
============================================================
DATA SPLITTING
============================================================
Training set:       4,000 samples
  - For training:   3,200 samples
  - For validation: 800 samples
Test set:           1,000 samples
```

```
# Creating a set without multicolinearity for linear models

print(f"\nFeature count:      {x_train.shape[1]} features")

correlated_feature = 'rolling_12mo_avg_basket'

x_train_clean = x_train.drop(columns=[correlated_feature])
x_test_clean = x_test.drop(columns=[correlated_feature])
x_train_split_clean = x_train_split.drop(columns=[correlated_feature])
x_val_clean = x_val.drop(columns=[correlated_feature])

print(f"\nClean feature count (linear models): {x_train_clean.shape[1]} feature
print(f"Removed feature: '{correlated_feature}'")
```

```
Feature count:      14 features

Clean feature count (linear models): 13 features
Removed feature: 'rolling_12mo_avg_basket'
```

```
# Converting to DMatrix for XGBoost
print("\nCreating DMatrix objects for XGBoost...")
dtrain = xgb.DMatrix(x_train_split, label=y_train_split)
dval = xgb.DMatrix(x_val, label=y_val)
dtest = xgb.DMatrix(x_test, label=y_test)
```

```
Creating DMatrix objects for XGBoost...
```

Model Training

Linear Regression Model

```
# Linear Regression Model
from sklearn.linear_model import LinearRegression

print("\nTraining Linear Regression...")
lr = LinearRegression()
lr.fit(x_train_split_clean, y_train_split)  # Train on log scale

# Predict and convert back to original scale
y_pred_lr_log = lr.predict(x_test_clean)
y_pred_lr = np.expm1(y_pred_lr_log)
y_test_actual = np.expm1(y_test)

# Calculate metrics
lr_r2 = r2_score(y_test_actual, y_pred_lr)
lr_mae = mean_absolute_error(y_test_actual, y_pred_lr)
lr_rmse = np.sqrt(mean_squared_error(y_test_actual, y_pred_lr))
```

```
Training Linear Regression...
```

ElasticNet Regression

```
# ElasticNet Regression
from sklearn.linear_model import ElasticNet

print("Training ElasticNet...")
enet = ElasticNet(alpha=0.001, l1_ratio=0.5, random_state=42)
enet.fit(x_train_split_clean, y_train_split)  # Train on log scale

# Predict and convert back
y_pred_enet_log = enet.predict(x_test_clean)
y_pred_enet = np.expm1(y_pred_enet_log)

# Calculate metrics
enet_r2 = r2_score(y_test_actual, y_pred_enet)
enet_mae = mean_absolute_error(y_test_actual, y_pred_enet)
enet_rmse = np.sqrt(mean_squared_error(y_test_actual, y_pred_enet))
```

```
Training ElasticNet...
```

XGBoost

```
print("Training XGBoost...")
# Setting the parameters
params = {
    "objective": "reg:squarederror",  # regression with squared error
```

```
    "max_depth": 3,
    "min_child_weight": 10,
    "eta": 0.03,                        # learning rate
    "subsample": 0.8,
    "colsample_bytree": 0.8,
    "alpha": 0.01,                      # L1 regularization
    "lambda": 0.1,                      # L2 regularization
    "eval_metric": "mae"                # monitoring metric
}

# Train with early stopping
evals = [(dtrain, "train"), (dval, "validation")]
bst = xgb.train(
    params,
    dtrain,
    num_boost_round=2000,
    evals=evals,
    early_stopping_rounds=10,
    verbose_eval=False
)

print(f"Best iteration: {bst.best_iteration}")

# Predict and convert back
y_pred_xgb_log = bst.predict(dtest)
y_pred_xgb = np.expm1(y_pred_xgb_log)

# Calculate metrics
xgb_r2 = r2_score(y_test_actual, y_pred_xgb)
xgb_mae = mean_absolute_error(y_test_actual, y_pred_xgb)
xgb_rmse = np.sqrt(mean_squared_error(y_test_actual, y_pred_xgb))
```

```
Training XGBoost...
Best iteration: 156
```

General Performance (Model Error)

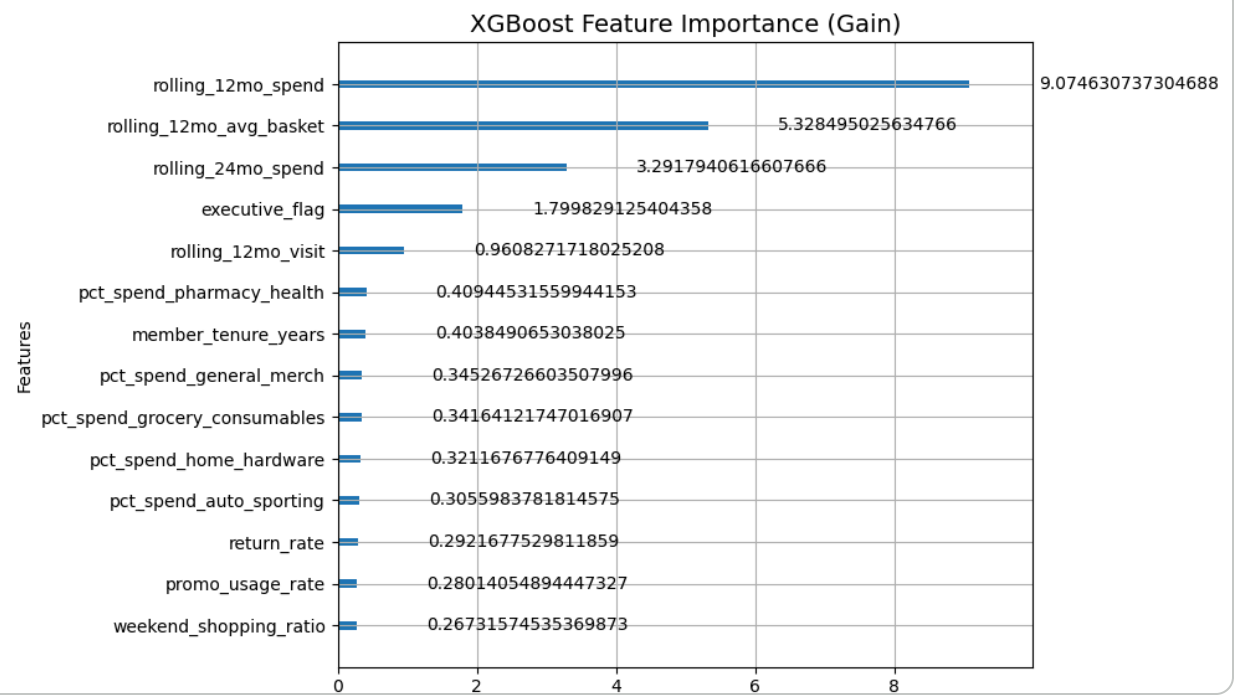
```
print("\n" + "="*60)
print("MODEL COMPARISON (TEST SET PERFORMANCE)")
print("="*60)
print(f"{'Model':<20} {'R²':<10} {'MAE':<15} {'RMSE':<15}")
print("-"*60)

models_metrics = [
    ("Linear Regression", lr_r2, lr_mae, lr_rmse),
    ("ElasticNet", enet_r2, enet_mae, enet_rmse),
    ("XGBoost", xgb_r2, xgb_mae, xgb_rmse)
]
```

```
for name, r2, mae, rmse in models_metrics:
    print(f"{name:<20} {r2:.4f}     ${mae:,.2f}       ${rmse:,.2f}")

print("="*60)

# Find best model
best_model_idx = np.argmax([lr_r2, enet_r2, xgb_r2])
best_model_name = ["Linear Regression", "ElasticNet", "XGBoost"][best_model_idx
print(f"\n🏆 Best Model: {best_model_name} (R² = {max(lr_r2, enet_r2, xgb_r2):
```

```
============================================================
MODEL COMPARISON (TEST SET PERFORMANCE)
============================================================
Model                R²         MAE             RMSE
------------------------------------------------------------
Linear Regression    0.6614     $264.66       $333.65
ElasticNet           0.6625     $264.42       $333.11
XGBoost              0.6864     $256.20       $321.07
============================================================

🏆 Best Model: XGBoost (R² = 0.6864)
```

Feature Importance

```
plt.figure(figsize=(10, 6))
xgb.plot_importance(bst, ax=plt.gca(), importance_type='gain')
plt.title('XGBoost Feature Importance (Gain)', fontsize=14)
plt.tight_layout()
plt.show()
```



SHAP

- SHAP is a way to help explain each prediction by showing how features push spending up or down.

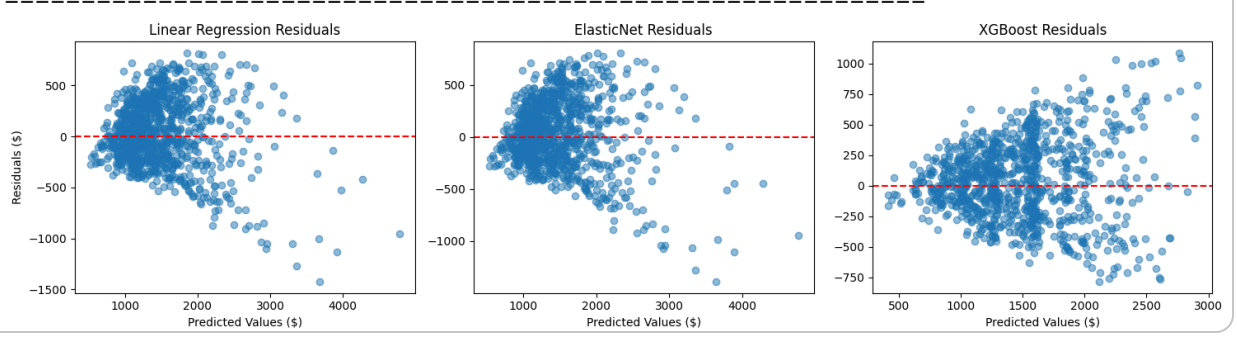
```
# SHAP Explainer
explainer = shap.Explainer(bst)
shap_values = explainer(dtest)

# Summary plot
plt.figure(figsize=(12, 6))
shap.summary_plot(shap_values, x_test, show=False)
plt.title('SHAP Feature Importance', fontsize=14)
plt.tight_layout()
plt.show()

# SHAP summary bar plot
plt.figure(figsize=(10, 6))
shap.summary_plot(shap_values, x_test, plot_type="bar", show=False)
plt.title('Mean Absolute SHAP Values', fontsize=14)
plt.tight_layout()
plt.show()
```

Show hidden output

Residual Analysis

```
print("\n" + "="*60)
print("RESIDUAL ANALYSIS")
print("="*60)

fig, axes = plt.subplots(1, 3, figsize=(15, 4))

# Linear Regression residuals
axes[0].scatter(y_pred_lr, y_test_actual - y_pred_lr, alpha=0.5)
axes[0].axhline(y=0, color='r', linestyle='--')
axes[0].set_xlabel('Predicted Values ($)')
axes[0].set_ylabel('Residuals ($)')
axes[0].set_title('Linear Regression Residuals')

# ElasticNet residuals
axes[1].scatter(y_pred_enet, y_test_actual - y_pred_enet, alpha=0.5)
axes[1].axhline(y=0, color='r', linestyle='--')
axes[1].set_xlabel('Predicted Values ($)')
axes[1].set_title('ElasticNet Residuals')

# XGBoost residuals
axes[2].scatter(y_pred_xgb, y_test_actual - y_pred_xgb, alpha=0.5)
axes[2].axhline(y=0, color='r', linestyle='--')
axes[2].set_xlabel('Predicted Values ($)')
axes[2].set_title('XGBoost Residuals')

plt.tight_layout()
plt.show()
```

```
============================================================
RESIDUAL ANALYSIS
============================================================
```



Individual Prediction Analysis

```
print("\n" + "="*60)
print("INDIVIDUAL PREDICTION EXAMPLE")
print("="*60)

# Select a random test case
```

```
sample_idx = 0

# full set
sample_features_full = x_test.iloc[sample_idx]
# Clean set for linear models
sample_features_clean = x_test_clean.iloc[sample_idx]

actual_value = np.expm1(y_test.iloc[sample_idx])

# Get predictions from all models
lr_pred = np.expm1(lr.predict(sample_features_clean.to_frame().T)[0])
enet_pred = np.expm1(enet.predict(sample_features_clean.to_frame().T)[0])
xgb_pred = np.expm1(bst.predict(xgb.DMatrix(sample_features_full.to_frame().T))

print(f"\nSample Member Features (full set):")
for feature, value in sample_features_full.items():
    print(f"  {feature}: {value: .4f}")

print(f"\nActual Holiday Spend: ${actual_value:,.2f}")
print(f"\nModel Predictions:")
print(f"  Linear Regression: ${lr_pred:,.2f} (Error: ${abs(actual_value - lr_pr
print(f"  ElasticNet:        ${enet_pred:,.2f} (Error: ${abs(actual_value - ene
print(f"  XGBoost:           ${xgb_pred:,.2f} (Error: ${abs(actual_value - xgb_

# Optional: Show which feature was removed for linear models
print(f"\nNote: Linear models used features without '{correlated_feature}'")
```

```
============================================================
INDIVIDUAL PREDICTION EXAMPLE
============================================================

Sample Member Features (full set):
  rolling_12mo_spend:  3582.7028
  rolling_12mo_visit:  24.0000
  rolling_12mo_avg_basket:  149.2793
  rolling_24mo_spend:  5408.5631
  pct_spend_grocery_consumables:  0.2828
  pct_spend_pharmacy_health:  0.1854
  pct_spend_home_hardware:  0.3070
  pct_spend_auto_sporting:  0.0473
  pct_spend_general_merch:  0.1774
  member_tenure_years:  13.3000
  return_rate:  0.0141
  executive_flag:  1.0000
  weekend_shopping_ratio:  0.5961
  promo_usage_rate:  0.1153

Actual Holiday Spend: $1,460.97

Model Predictions:
  Linear Regression: $1,158.06 (Error: $302.91)
  ElasticNet:        $1,151.33 (Error: $309.63)
  XGBoost:           $1,190.66 (Error: $270.31)
```

```
Note: Linear models used features without 'rolling_12mo_avg_basket'
```

Summary and Recommendations

```
print("\n" + "="*60)
print("SUMMARY AND RECOMMENDATIONS")
print("="*60)

print("\n1. DATA PREPARATION:")
print("   ✓ Applied log transformation to handle skewness")
print("   ✓ All models trained on consistent log-transformed scale")
print("   ✓ Proper train/validation/test splitting")
print("   ✓ Created separate feature sets for different model types")

print("\n2. CORRELATED FEATURES HANDLING:")
print(f"   ⚠️  Found highly correlated pair: rolling_12mo_spend ↔ rolling_12mo_
print("   Implemented model-specific feature engineering:")
print("     ✓ XGBoost: Retained both features (robust to multicollinearity)")
print("     ✓ Linear/ElasticNet: Removed rolling_12mo_avg_basket (avoids coeff

print("\n3. MODEL PERFORMANCE:")
print(f"   Best performing model: {best_model_name} (R² = {max(lr_r2, enet_r2,
print(f"   Linear Regression: R² = {lr_r2:.4f}, MAE = ${lr_mae:.2f}")
print(f"   ElasticNet:        R² = {enet_r2:.4f}, MAE = ${enet_mae:.2f}")
print(f"   XGBoost:           R² = {xgb_r2:.4f}, MAE = ${xgb_mae:.2f}")

print("\n4. KEY INSIGHTS:")
print("   ✓ Spending history (rolling_12mo_spend) is strongest predictor")
print("   ✓ Tree-based and linear models require different feature engineering
print("   ✓ Residual patterns similar across models (no systematic bias)")
print("   ✓ Executive membership status significantly impacts spending")

print("\n5. RECOMMENDATIONS:")
print(f"   Production: Deploy {best_model_name} for holiday spend predictions")
print("   Feature engineering: Maintain separate preprocessing pipelines for:")
print("     - Tree models: Use all available features")
print("     - Linear models: Remove highly correlated features")
print("   Monitoring: Track model drift using MAE on validation data")
print("   Business: Focus on members with high annual spend for holiday campaig
```

```
============================================================
SUMMARY AND RECOMMENDATIONS
============================================================

1. DATA PREPARATION:
   ✓ Applied log transformation to handle skewness
   ✓ All models trained on consistent log-transformed scale
   ✓ Proper train/validation/test splitting
```